Domenic J. Jernigan

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EDUCATION

University of California, San Diego — La Jolla, CA Expected Jun 2027
B.S. Cognitive Science, Specialization in Machine Learning & Neural Computation | Minor in Computer Science | GPA: 3.56

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, JSON, Typescript
ML & Data: pandas, NumPy, scikit-learn, Jupyter, Tableau, SQL Workbench, Microsoft Excel
Dev Tools: Git/GitHub, GDB, Valgrind, Bash, Vim, VS Code, Vercel, React, Next.js Framework

WORK EXPERIENCE

Pedagogical Learning Apprentice — UC San Diego, COGS 108: Data Science in Practice Apr 2026 – Present

- Competitively selected by course faculty to support instruction in applied machine learning and data science across weekly sections and project reviews.
- Will mentor students through end-to-end Python pipelines using pandas and scikit-learn — regression, classification, time-series forecasting, and unsupervised clustering
- Debug student code, hold office hours, and advise research-style group projects from problem scoping through model evaluation.

ML Research Project Lead — UC San Diego, COGS 108: Data Science in Practice Jan 2026 – Mar 2026

- Led a peer team on an end-to-end ML study classifying Bitcoin as a financial instrument, extending peer-reviewed literature on cryptocurrency market behavior; BTC reached correlations ~0.60 - 0.70 with S&P 500 Index around Covid-era
- Engineered features and trained regression, classification, and clustering models in scikit-learn; applied time-series forecasting to benchmark Bitcoin against equities, commodities, and currency indices
- Owned scoping, task delegation, and experiment design across a 10-week project; authored the final report and presentation — earned an A in the course

Data Operations Specialist — UC San Diego, Office of Undergraduate Admissions May 2025 – Present

- Process high-volume applicant records — transcripts, test scores, correspondence — through scanning, indexing, and validation workflows in Slate CRM and OnBase
- Maintain data integrity and FERPA compliance across Slate, OnBase, and Blade; reconcile discrepancies flagged during peak admissions cycles
- Collaborate with Enrollment Management on workflow improvements supporting peak-cycle processing and continuous pipeline refinement

VP of Finance & Operations — Interfraternity Council, UC San Diego Dec 2024 – Jan 2026

- Managed budgets, fund allocation, and expense tracking for a Tier-5 council overseeing 15 fraternity chapters and hundreds of members
- Built financial reports that improved transparency of dues, reimbursements, and multi-event budgeting across the organization • Partnered with executive board and chapter leaders on compliance reviews and quarterly financial strategy

PROJECTS

AI Log Triage Agent — Apr 2026 – Present

- Python, Anthropic API Built and evaluated a CLI tool that uses Claude to summarize log incidents and propose root causes. Implemented a regex-based clustering layer to reduce LLM input tokens by 80–90% on synthetic logs, and validated output quality against a 20-case hand-labeled eval set (87% severity-classification agreement). ~150 LOC; processes 10K-line logs in <8s at ~$0.005/run

Pioneer Shell — Custom Unix-like Shell: Mar 2025
Virtual Memory Allocator — Custom malloc/free: Feb 2025